

ProofStack

Work > Words — Evidence-First Hiring

A verified track record of real engineering work — replacing resume claims with build-tested, reviewed, and AI-defended proof of skill.

TEAM ZENITH · HM26-7096

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College(s): [Add college name(s)]

SUB-PROBLEM: VERIFICATION

Hiring Runs on Claims, Not Proof.

Sub-problem: Verification — verified proof-of-skill for technical hiring.

- A resume is a claim, not evidence — there's no proof behind the bullet points.
- Recruiters filter blind, screening on pedigree signals instead of demonstrated work.
- Skilled-but-uncredentialed talent stays invisible to the hiring pipeline.
- No way to tell real, original work apart from copied or mocked work.

73%

of technical hires fail to match their self-reported skill level

Three Roles, One Connected Platform.



Developer / Student

Builds and submits real projects. Proves skill through evidence — not resume claims — even with zero prior projects on record.



Recruiter

Screens fast and needs an explainable shortlist backed by verified signals, not self-reported skills.



Institution

Bulk-onboards and vouches for its own students under an authorized organization account.

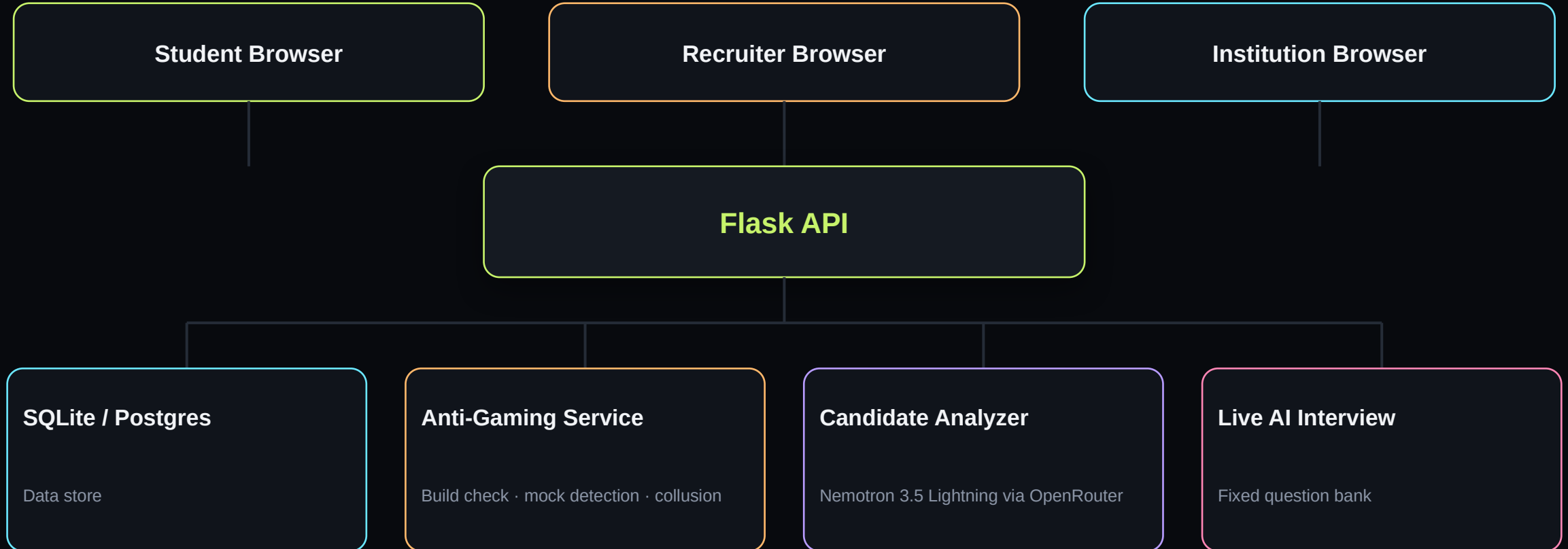
BuildProof: One Connected Loop.

The same data model carries every user type end to end — no disconnected dashboards.



Build → Document → Review → Rank → Discover → Hire — evidence flows through the same pipeline recruiters search, so there's nothing to reconcile between “what the student did” and “what the recruiter sees.”

System Architecture.



Frontend: HTML/CSS/JS · Backend: Python Flask · Build execution: simulated runner (MVP) → Docker/gVisor (prod)

The Five Hard Constraints.

Constraint	Our approach	Status
User Types	3 end-to-end roles: Student, Recruiter, Institution	✓
Core Entities	7 entities, one data model — Ranking is derived, not stored	✓
Ranking Logic	Proof Score = Base × Difficulty × Recency × Breadth, fully explainable	✓
Prevent Gaming	5 mechanisms: build check, mock detection, collusion resistance, cold-start, integrity events	✓
Offline Mode	Not yet supported — platform assumes connectivity end-to-end	✗

AI Pre-Screens. Humans Decide.

Our approach: a live AI interview feeds a mandatory human-reviewed verdict. Rejected alternative: a fully-automated AI pass/fail.

Dimension	Our approach — AI + human review	Rejected — fully-automated verdict
Accountability	A human expert makes the final call	AI verdict alone, no appeal path
Recruiter trust	Explainable, defensible decisions	Black-box risk on hiring calls
Speed	Slower — needs reviewer bandwidth	Instant
Build effort (hackathon)	Needs a reviewer workflow to ship	Simpler to build in 72h

Cost accepted: we trade instant, fully-automated verdicts for slower turnaround and reviewer bandwidth — in exchange for accountability recruiters can trust.

Three Flows, Traced End to End.

What the system actually does with a submission — the good case, the caught case, and the honest gap.

Normal flow

- 1 Student submits a solution
- 2 Build passes
- 3 Proof score updates live

Proves: a clean submission flows straight through to ranking.

Flagged input

- 1 Student submits a solution
- 2 GitHub-mocking check fails
- 3 Routed to human review — never auto-rejected

Proves: gaming attempts are caught, but a human still decides.

Offline state

- 1 Student opens app with no connectivity
- 2 No offline queue yet
- 3 Submission blocked — on the roadmap

Proves: a disclosed gap, not a hidden one (see Limits & Scale).

What Breaks at City Scale.

What breaks first	Why	Fix
Simulated build runner	Any real concurrent load	Isolated Docker / gVisor execution + job queue
Reviewer bandwidth	Doesn't scale linearly with submission volume	ML-assisted reviewer calibration and weighting
Duplicate / near-duplicate work	Manual GitHub heuristics get harder to police at scale	Automated dedup + similarity pipeline

Known gaps

No offline submission mode · GitHub-mocking detection is heuristic only (repo age vs. commit count) · No webcam gaze-tracking or keystroke-identity biometrics — a deliberate scope decision, not an oversight (see docs/limitations.md).

Roadmap.

1

Pilot with one institution

Onboard one college's cohort end-to-end to stress-test the vouching flow.

2

Real build runner

Replace the simulated flag with isolated Docker / gVisor execution.

3

Recruiter marketplace expansion

Open verified-talent search to more hiring partners beyond the pilot.

AI disclosure

Claude (Anthropic) was used for development scaffolding. In-product AI: NVIDIA Nemotron 3.5 Lightning via OpenRouter powers the live AI interview and candidate analyzer, with a deterministic offline fallback. Full detail in ai.md.

Live MVP: [Add URL after deployment]

Repo: github.com/LEHANYA2023/Proofstack1-HM26-7096

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